

A Literature-Implied Hilbert-Gap Obstruction for Remark 6.11

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Status

This is a literature-implied partial-subcase answer. It records that a later theorem of Johnson and Schechtman rules out the Hilbert-gap versions of Mathieu–Tradacete Remark 6.11. It is not a full answer to the all-exponent approximate factorization question.

Source Question

Mathieu and Tradacete ask in Remark 6.11 of [1] whether every $T \in \mathcal{S}(L_p[0, 1])$, $1 < p < \infty$ and $p \neq 2$, is approximately (ℓ_r, ℓ_s) -factorizable for some $r < s$. They also ask the stronger finite-dimensional-sum version.

then X should contain a subspace isomorphic to ℓ_p , $\ell_{p'}$ or c_0 (although this last option is impossible because of the weak compactness of $L_A R_B$). Hence, the case $X = \ell_s$ as above can also be ruled out by this approach.

Remark 6.11. We do not know whether every $T \in \mathcal{S}(L_p)$ is approximately (ℓ_r, ℓ_s) -factorizable for some $r < s$. In fact, we do not know whether for an operator $T \in \mathcal{S}(L_p)$, and every $\varepsilon > 0$, there exist $r(\varepsilon) < s(\varepsilon)$, two sequences of finite dimensional spaces $(X_n)_{n \in \mathbb{N}}$, $(Y_n)_{n \in \mathbb{N}}$ and operators

$$\begin{aligned} T_1^\varepsilon &: L_p \longrightarrow (\oplus X_n)_{\ell_{r(\varepsilon)}} \\ T_2^\varepsilon &: (\oplus X_n)_{\ell_{r(\varepsilon)}} \longrightarrow (\oplus Y_n)_{\ell_{s(\varepsilon)}} \\ T_3^\varepsilon &: (\oplus Y_n)_{\ell_{s(\varepsilon)}} \longrightarrow L_p \end{aligned}$$

such that $\|T - T_3^\varepsilon T_2^\varepsilon T_1^\varepsilon\| \leq \varepsilon$. Keeping in mind the previous comment, by Remark 5.3 and the argument in the proof of Theorem 6.9, such a factorization would yield an alternative direct proof of [15, Theorem 2.9].

In connection with approximate factorization the following property of strictly

Supporting Literature

Let $\Gamma_2(L_p)$ denote the ideal of operators on L_p that factor through a Hilbert space, and let $\overline{\Gamma_2}(L_p)$ be its norm closure. Johnson and Schechtman prove in [2], Remark 6, that their large closed ideals in $L(L_p)$ do not contain the strictly singular ideal. In the introduction they explicitly draw the consequence that $\overline{\Gamma_2}(L_p)$ does not contain all strictly singular operators on L_p , for every $1 < p \neq 2 < \infty$.

Thus, for every $1 < p \neq 2 < \infty$, there exists

$$S \in \mathcal{S}(L_p) \setminus \overline{\Gamma_2}(L_p).$$

Identification

If $1 < p < 2$ and an operator T is approximately (ℓ_p, ℓ_2) -factorizable, then T is a norm limit of operators

$$L_p \xrightarrow{A_\varepsilon} \ell_p \xrightarrow{B_\varepsilon} \ell_2 \xrightarrow{C_\varepsilon} L_p.$$

Each approximant factors through the Hilbert space ℓ_2 , namely as $C_\varepsilon(B_\varepsilon A_\varepsilon)$. Hence $T \in \overline{\Gamma_2}(L_p)$.

Similarly, if $2 < p < \infty$ and T is approximately (ℓ_2, ℓ_p) -factorizable, then every approximant

$$L_p \xrightarrow{A_\varepsilon} \ell_2 \xrightarrow{B_\varepsilon} \ell_p \xrightarrow{C_\varepsilon} L_p$$

factors through ℓ_2 , so again $T \in \overline{\Gamma_2}(L_p)$.

Combining this elementary ideal observation with Johnson–Schechtman gives:

- $1 < p < 2$: some $S \in \mathcal{S}(L_p)$ is not approximately (ℓ_p, ℓ_2) -factorizable,
- $2 < p < \infty$: some $S \in \mathcal{S}(L_p)$ is not approximately (ℓ_2, ℓ_p) -factorizable.

Scope and Remaining Frontier

This does not disprove the full assertion in Remark 6.11. A strictly singular operator outside $\overline{\Gamma_2}(L_p)$ might still be approximately (ℓ_r, ℓ_s) -factorizable for a non-Hilbert pair $r < s$ with neither exponent equal to 2. The result only shows that the natural Hilbert-gap upgrade of the DFJP/Johnson factorization machinery cannot solve the problem for all strictly singular operators.

The supporting authors were not, in this paper, explicitly answering Mathieu–Tradacete Remark 6.11. The relation above is an agent-identified implication from their closed-ideal theorem.

Search Evidence

Run-index searches for 1810.09194, approximate factorization, and Schreier/strictly singular keywords found the prior conditional and $\mathcal{S}_1$ packets for this target. Web searches for the exact phrases “approximately (ℓ_r, ℓ_s) -factorizable” and variants did not find a later direct answer to Remark 6.11. The Johnson–Schechtman paper arXiv:2003.11414 was found through closed-ideal searches around $L(L_p)$, strictly singular operators, and Γ_2 .

References

- [1] M. Mathieu and P. Tradacete, *Strictly singular multiplication operators on $\mathcal{L}(X)$* , arXiv:1810.09194, 2018.
- [2] W. B. Johnson and G. Schechtman, *The number of closed ideals in $L(L_p)$* , arXiv:2003.11414, 2020.